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SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Programme: **B.E / B.Tech**

Course Code: **MLA0204**

Course Name: **FUNDAMENTALS OF MACHINE LEARNING**

Duration: **1:30 Hr**

Max Marks: **50 Marks**

Date : 31/10/26

ASSIGNMENT – II

UNIT II – SUPERVISED LEARNING

Topics: *Linear regression and classification, naïve Bayes, discriminant functions, logistic regression, decision trees and pruning, neural networks and back-propagation, SVM, ensembles (bagging, boosting)*

Course Outcome

On successful completion of this assignment, students will be able to:

CO	Course Outcome Statement
CO2	Apply supervised learning algorithms for regression, classification, and predictive modeling tasks to solve structured data problems in practical applications.

Answer all the Tasks

Task 1. The following data relate the number of training epochs x to a validation score y.

x	1	2	3	4	5
y	2.1	3.9	6.2	7.8	10.1

(a) Fit $y = w_0 + w_1x$ by ordinary least squares using the normal equations. (b) Predict y at $x = 6$. (c) Compute the residual sum of squares, MSE and R^2. (d) Analyse the reliability of predicting at $x = 6$.

(5) (CO2) (BL4)

Task 2. For mean-centred data with $Sxx = \Sigma x^2 = 10$ and $Sxy = \Sigma xy = 19.9$ (single feature, no intercept needed). compute the ridge-regression slope for $\lambda = 5$. Calculate the percentage shrinkage relative to the least-squares slope and explain the bias–variance effect of increasing λ.

(2) (CO2) (BL4)

Task 3. An e-mail filter uses two binary features: Offer (contains the word 'offer') and Link (contains a hyperlink). Ten labelled e-mails are available.

#	Offer	Link	Class
1	Y	Y	Spam
2	Y	N	Spam
3	Y	Y	Spam
4	N	Y	Spam
5	Y	Y	Spam

#	Offer	Link	Class
6	N	N	Ham
7	N	Y	Ham
8	Y	N	Ham
9	N	N	Ham
10	N	N	Ham

(a) Estimate the priors and the class-conditional probabilities without smoothing. (b) Classify an e-mail with Offer = Y, Link = Y and report the posterior P(Spam | x). (c) Repeat (b) with Laplace smoothing (α = 1). (d) Classify an e-mail with Offer = N, Link = N (with smoothing) and analyse the effect of the conditional-independence assumption and of smoothing.

(6) (CO2) (BL4)

Task 4. A logistic-regression classifier $p = \sigma(w_1x_1 + w_2x_2 + b)$ is initialised with $w = (0, 0)$, $b = 0$ and trained on two samples: $x^{(1)} = (1, 2)$ with $y = 1$, and $x^{(2)} = (2, 1)$ with $y = 0$. Perform one batch gradient-descent step with learning rate $\eta = 0.5$ on the mean cross-entropy loss. Report the gradient, the updated parameters, the predicted probabilities and the loss before and after the update, and analyse the result.

(4) (CO2) (BL4)

Task 5. A node contains 10 training samples (6 Yes, 4 No). A candidate attribute A splits it into $A = a_1$ (4 samples: 4 Yes, 0 No) and $A = a_2$ (6 samples: 2 Yes, 4 No). Calculate the entropy of the node, the information gain of the split and the Gini-impurity decrease, and evaluate whether the split is worthwhile.

(3) (CO2) (BL4)

Task 6. In a decision tree trained on 100 samples, an internal node t with a four-leaf subtree T_t makes 3 training errors, whereas collapsing t into a single leaf would make 8 errors. Using cost-complexity pruning with $R_\alpha(T) = R(T) + \alpha|\text{leaves}|$, compute the critical value α at which the subtree and the single leaf have equal cost, and evaluate whether the subtree should be pruned for $\alpha = 0.01$ and $\alpha = 0.02$.

(3) (CO2) (BL5)

Task 7. A network has two inputs, one sigmoid hidden unit and one sigmoid output unit (no biases): $h = \sigma(w_1x_1 + w_2x_2)$, $\hat{y} = \sigma(vh)$, with $w_1 = 0.4$, $w_2 = 0.3$, $v = 0.5$. For the input $x = (1, 2)$ with target $t = 1$ and squared-error loss $E = \tfrac{1}{2}(t - \hat{y})^2$, perform one forward pass and one back-propagation update with $\eta = 0.5$. Report h, $\hat{y}$, E, the error terms δ, the gradients, the updated weights and the new loss.

(4) (CO2) (BL4)

Task 8. A linear SVM in two dimensions has $w = (0.5, 0.5)$ and $b = -2$, with the support vectors (3, 3) (class +1) and (1, 1) (class −1). (a) Verify that the support vectors lie on the margin hyperplanes and compute the geometric margin width. (b) Classify the points (4, 2) and (1, 2) and compute their distances from the decision boundary. (c) Compute the hinge loss for (1, 2) with true label −1 and for (2, 2) with true label +1, and state which points violate the margin.

(4) (CO2) (BL4)

Task 9. Independent classifiers, each with accuracy 0.70, are combined by majority voting. Calculate the ensemble accuracy for 3 and 5 classifiers, comment on the gain from adding more classifiers, and analyse what happens to this gain if the classifiers' errors are strongly correlated.

(3) (CO2) (BL4)

Task 10. AdaBoost starts with five training samples of equal weight. The first weak learner misclassifies samples 2 and 5. Compute the weighted error, the learner weight α, the un-normalised and normalised weights after the update, and analyse how boosting changes the focus of the next learner.

(4) (CO2) (BL4)

Task 11. Two classes share the covariance matrix $\Sigma = [[2, 0.5],[0.5, 1]]$ and have means $\mu_1 = (1, 2)^T$ and $\mu_2 = (3, 1)^T$. (a) Derive the discriminant direction $w = \Sigma^{-1}(\mu_2 - \mu_1)$ and the decision threshold for equal priors. (b) Classify $x = (2.5, 1)$. (c) Evaluate whether the decision changes if the priors are $P(\omega_1) = 0.8$ and $P(\omega_2) = 0.2$.

(4) (CO2) (BL4)

Task 12. A bank evaluates three classifiers on 1,000 test loans (100 actual defaulters). Default is the positive class.

Model	TP	FN	FP	TN
A	70	30	90	810
B	85	15	180	720
C	60	40	30	870

A missed defaulter (FN) costs ₹50,000 and a wrongly rejected good applicant (FP) costs ₹5,000.

(a) Compute accuracy, precision, recall and F1-score for each model. [3]

(b) Compute the total expected cost of each model. [2]

(c) Critically evaluate which model should be deployed and explain why accuracy and F1 lead to a different choice than cost. [2]

(d) Propose two modelling or threshold-tuning steps to improve the chosen model and one validation safeguard against over-fitting. [1]

(8) (CO2) (BL6)

Assignment - 11

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② Task 2 - Linear Regression

Given;

x	1	2	3	4	5
y	2.1	3.9	6.2	7.8	10.1

a) Fit $y = w_0 + w_1 x$

Using the normal equations:

$$w_1 = \frac{n\Sigma xy - \Sigma x \Sigma y}{n\Sigma x^2 - (\Sigma x)^2} = 1.99$$

$$w_0 = \bar{y} - w_1 \bar{x} = 0.05$$

Therefore, $y = 0.05 + 1.99x$

b) Prediction at $x = 6$:

$$y = 0.05 + 1.99(6) = 11.99$$

c) Performance:

Residuals:

$[0.06, -0.13, 0.18, -0.21, 0.10]$

$RSS = 0.107$

$$MSE = \frac{0.107}{5} = 0.02124$$

$R^2 = 0.9973 \approx 99.73\%$

d) Reliability.

The R^2 value is very high, so the linear model fits the given data well. However, $x = 6$ is outside the training range (1-5), so the prediction is an extrapolation and is less reliable than predictions

(2) Task 2 - Ridge Regression

Given;

$S_{xx} = 10,\ S_{xy} = 19.9,\ \lambda = 5$

Ridge slope:

$$w_1 = \frac{S_{xy}}{S_{xx} + \lambda} = \frac{19.9}{15} = 1.3267$$

Least squares slope:

$$w_{LS} = \frac{19.9}{10} = 1.99$$

Percentage shrinkage:

$$\frac{1.99 - 1.3267}{1.99} \times 100 = 33.33\ \%$$

Effect:

increasing λ increases bias but reduce variance, making the model less sensitive to noise and reducing overfitting.

(3) Task 3 - Naive Bayes

There are 5 spam and 5 Ham emails

$P(\text{Spam}) = P(\text{Ham}) = 0.5$

Without smoothing:

- Spam:

$P(\text{offer}) = Y/\text{Spam}) = 4/5$

$P(\text{Link}) = Y/\text{Spam}) = 4/5$

- Ham:

$P(\text{offer} = Y/\text{HAM}) = 1/5$

$P(\text{Link} = Y/\text{HAM}) = 1/5$

For offer = Y, Link = Y.

Spam score:

$0.5 (0.8)(0.8) = 0.32$

Ham score:

$0.5 (0.2)(0.2) = 0.02$

Therefore

$$P(\text{Spam}/x) = \frac{0.32}{0.32+0.02} = 0.9412$$

(c) Laplace smoothing ($\alpha = 1$):

$$P(Y/\text{Spam}) = \frac{4+1}{5+2} = \frac{5}{7}$$

$$P(Y/\text{Ham}) = \frac{1+1}{5+2} = \frac{2}{7}$$

Thus,

$$P(\text{Spam}/Y,Y) = \frac{0.5 (5/7)^2}{0.5 (5/7)^2 + 0.5 (2/7)^2} = 0.8621$$

(d) For offer = N, Link = N with smoothing:

$P(N/\text{Spam}) = 2/7$, $P(N/\text{Ham}) = 5/7$

$P(\text{Spam}/N,N) = 0.20$

Hence the email is classified as Ham.

Analysis:

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Task 4 - Logistic Regression

Initial:

$w = (0,0)$, $b = 0$

Therefore both predicted probabilities are:

$p_1 = p_2 = 0.5$

For mean cross-entropy:

$\nabla w = \frac{1}{2} \sum (p - y) x$

Gradient:

$\nabla w = (0.25, -0.25)$

$\nabla b = 0$

With $\eta = 0.5$:

$w_{new} = w - 0.5 \nabla w$

$w = (-0.125, 0.125)$, $b = 0$

- **New probabilities:**

$p_1 = 0.5312$, $p_2 = 0.4688$

- **Loss before:**

0.6931

- **Loss after:**

0.6326

Analysis:

The loss decreased, showing that the gradient-descent update moves the classifier in the correct direction.

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Task 5) Decision tree

Parent node = 6 Yes, 4 No

- Entropy:

$$H = -0.6 \log_2(0.6) - 0.4 \log_2(0.4)$$

$$H = 0.971$$

- For A = a1: entropy = 0

- For A = a2:

$$H(a_2) = (-2/6)\log_2(2/6) - (4/6)\log_2(4/6) = 0.918$$

- Weighted entropy:

$$\frac{4}{10}(0) + \frac{6}{10}(0.918) = 0.551$$

- Information gain:

$$IG = 0.971 - 0.551 = 0.420$$

- Parent Gini:

$$G = 1 - (0.6)^2 - (0.4)^2 = 0.48$$

- Split Gini:

$$0.4(0) + 0.6(0.4444) = 0.2667$$

- Gini decrease:

$$0.48 - 0.2667 = 0.2133$$

- Conclusion:

The split is worthwhile because it gives a significant reduction in impurity.

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6 Task 6 Cost Complexity Pruning

- **Subtree Cost:**

$R_\alpha(T) = 3 + 4\alpha$

- **Single leaf Cost:**

$8 + \alpha$

At the Critical Value:

$3 + 4\alpha = 8 + \alpha$

$3\alpha = 5$

$\alpha = 1.667$

For:

$\alpha = 0.01, 0.02$

the subtree has lower Cost than the single leaf.

Conclusion:

The subtree should not be pruned for either value.

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7 Task 7 - Neural Network & Back Propagation

- **Input:**

$x = (1, 2)$

- **Hidden activation:**

$z_h = 0.4(1) + 0.3(2) = 1$

$h = \sigma(1) = 0.7311$

- **Output:**

$z_o = 0.5(0.7311) = 0.3655$

$\hat{y} = 0.5904$

Error:

$$E = \frac{1}{2}(1 - 0.5904)^2 = 0.0839$$

Output error term:

$$\delta_o = -0.0991$$

Hidden error term:

$$\delta_h = -0.00974$$

Gradients:

$$\frac{\partial E}{\partial v} = -0.072442$$

$$\frac{\partial E}{\partial w_1} = -0.00974$$

$$\frac{\partial E}{\partial w_2} = -0.01948$$

Updated weights:

$$v = 0.5362$$

$$w_1 = 0.4049$$

$$w_2 = 0.3097$$

New loss:

$$E_{new} = 0.0811$$

The loss decreased, so the update improved the network.

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(8) Task 8 – SVM

Given;

$$w = (0.5, 0.5),\ b = -2$$

Decision function:

$$f(x) = 0.5x_1 + 0.5x_2 - 2$$

(a) Support vectors:

- For (3,3): $f = 1$ ✓
- For (1,1): $f = -1$ ✓

Therefore, both lie on the margin hyperplanes.

$$\|w\| = \sqrt{0.5^2 + 0.5^2} = 0.7071$$

- Margin width:

$$\frac{2}{\|w\|} = 2.828$$ ✓

(b) Classification:

- For (4,2): $f = 1$

- Class = +1, distance:

1.414 ✓

- For (1,2): $f = -0.5$

- Class = −1, distance:

0.707 ✓

(c) Hinge loss

For (1,2), $y = -1$:

$$L = \max(0, 1 - yf) = 0.5$$

- For (2,2), $y = +1$:

$f = 0$ ✓

$L = 1$

Both points violate the margin.

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Task 9 - Ensemble Majority Voting

Individual accuracy:

P = 0.70

For 3 classifiers:

Majority means at least 2 correct

$$P = \binom{3}{2}(0.7)^2(0.3) + (0.7)^3$$

$$P = 0.784 = 78.4\%$$ 1

- For 5 classifiers:

Majority means at least 3 correct

$$P = \binom{5}{3}(0.7)^3(0.3)^2 + \binom{5}{4}(0.7)^4(0.3) + (0.7)^5$$

$$P = 0.8369 = 83.69\%$$ 1

Adding classifiers improves accuracy, but the improvement becomes smaller.

If classifiers errors are strongly correlated, they tend to make the same mistakes, so majority voting provides much less benefit. 1

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10) Task 10 - AdaBoost

- Five samples have equal weights
  $w_i = 0.2$
  two samples are misclassified.
- Weighted error:
  $\epsilon = 2(0.2) = 0.4$
- Learner weight:
  $\alpha = \frac{1}{2} \ln \frac{1-\epsilon}{\epsilon}$
  $= \frac{1}{2} \ln \frac{0.6}{0.4} = 0.2027$
- For misclassified samples
  $w' = 0.2 e^{0.2027} = 0.2449$
- For correctly classified samples
  $w' = 0.2 e^{-0.2027} = 0.16330$
- Normalization gives:
  - samples 2 and 5: 0.25 each
  - other 3 samples: 0.167 each
- Analysis:

AdaBoost increases the weights of incorrectly classified samples, forcing the next learner to focus more on difficult examples.

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Discriminant Function

$$\Sigma = \begin{bmatrix} 2 & 0.5 \\ 0.5 & 1 \end{bmatrix}$$

$\mu_1 = (1, 2)^T$, $\mu_2 = (3, 1)^T$

Difference:

$\mu_2 - \mu_1 = (2, -1)^T$

$$\Sigma^{-1} = \begin{bmatrix} 0.5714 & -0.2857 \\ -0.2857 & 1.1429 \end{bmatrix}$$

Therefore,

$w = \Sigma^{-1}(\mu_2 - \mu_1)$

$w = (1.4286, -0.7143)$

Equal prior threshold:

0.2857

For $x = (2.5, 1)$:

$w^T x = 1.4286(2.5) - 0.7143(1) = 2.8571$

Since $2.8571 > 0.2857$

$x \rightarrow w_2$

If for $P(w_1) = 0.8$, $P(w_2) = 0.2$ the threshold becomes:

$0.2857 - \ln(0.2/0.8) = 1.6720$

Since $2.8571 > 1.6720$, the decision still remains w_2

(12) Task 12 Model Evaluation

The gives 1,000 test loans and the TP, FN, FP and TN values for models A, B and C

(a) Accuracy Precision Recall and F1

Model	Accuracy	Precision	Recall
A	88.0%	43.75%	70.0%
B	80.5%	32.08%	85.0%
C	93.0%	66.67%	60.0%

(b) Total Expected Cost

FN Cost = £50,000

FP Cost = £5,000

Cost = FN (50,000) + FP (5,000)

Model A:

30 (50,000) + 90 (5,000) = £19,50,000

Model B:

15(50,000) + 180 (5,000) = £16,50,000

Model C:

40 (50,000) + 30 (5,000) = £21,50,000

(c) Best model

Based only on accuracy and F1, Model C looks best

at the business cost says model B is best, because it has the lowest total cost:

₹ 16,30,000

The reason is that a missed defaulter's costs ₹50,000, which is 10 times the cost of a false positive therefore, reducing FN is more important than maximizing accuracy

5) Improvements

1. Tune the classification threshold: to improve recall and reduce costly false negatives.

2. Use class weighted learning: so that default cases receive higher importance.

3. Validation safeguard: use stratified cross-validation and evaluate performance on a separate validation/ test set to detect overfitting.

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ANSWER SCRIPT EVALUATION SHEET

ASSIGNMENT - II

Programme	B.E / B.Tech		
Course Name	**Fundamentals of Machine learning.**	**Course Code**	**MLA02**
Duration	1 Hr 30 Min	**Date**	31/10/26
		Max Marks	50

Register No.: 192525412

TASK	Max Marks	Marks Awarded
Task 1	5	
Task 2	2	
Task 3	6	
Task 4	4	
Task 5	3	
Task 6	3	
Task 7	4	
Task 8	4	
Task 9	3	
Task 10	4	
Task 11	4	
Task 12	8	
TOTAL	**50**	

Total in words: